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Estimating dynamic equilibrium models with stochastic volatility

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ABSTRACT

This paper develops a particle filtering algorithm to estimate dynamic equilibrium models with stochastic volatility using a likelihood-based approach. The algorithm, which exploits the structure and profusion of shocks in stochastic volatility models, is versatile and computationally tractable even in large-scale models. As an application, we use our algorithm and Bayesian methods to estimate a business cycle model of the US economy with both stochastic volatility and parameter drifting in monetary policy. Our application shows the importance of stochastic volatility in accounting for the dynamics of the data.

1. Introduction

This paper develops a particle filtering algorithm to estimate dynamic equilibrium models with stochastic volatility using a likelihood-based approach. The novelty of our algorithm is that it does not require the presence of linear measurement errors to evaluate the likelihood function of the model. In order to do that, we characterize the properties of the solution of these models when approximated with the second-order expansion. As an application of our procedure, we estimate a medium-size business cycle economy.

Our results are useful because, motivated by the findings of Stock and Watson (2002) and Sims and Zha (2006), many recent papers have built dynamic equilibrium models with volatility shocks (also known as uncertainty shocks). Among them, we can highlight Fernández-Villaverde and Rubio-Ramirez (2007), Justiniano and Primiceri (2008), Bloom (2009), and Fernández-Villaverde et al. (2010b). In these models, and in the tradition of stochastic volatility (Shephard, 2008), there are two types of shocks: structural shocks (shock to productivity, to preferences, etc.) and volatility shocks (shocks to the standard deviation of the innovations to the structural shocks).

To fulfill the promise in this literature, we need tools to estimate this class of models. However, the task is complicated by the inherent non-linearity that stochastic volatility generates. Linearization is ill-equipped to handle time-varying volatility because it yields certainty-equivalent policy functions. That is, volatility influences neither the agents' decision rules nor the laws of motion of the aggregate variables. Hence, to consider how stochastic volatility affects those factors, it is imperative to employ at least the second-order approximation to the equilibrium dynamics of the economy and to use simulation-based estimators of the likelihood.

To accomplish that last task, one could, in principle, rely on the baseline particle filter presented in Fernández-Villaverde and Rubio-Ramirez (2007). Unfortunately, that version of the particle filter requires, when estimating models with stochastic volatility, the presence of linear measurement errors in observables. Otherwise, we would be forced to solve a large quadratic system of equations with multiple solutions, an endeavor for which there are no suitable algorithms. Although measurement errors are plausible, they complicate identification in small samples and entangle the interpretation of the results.

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To get around this problem, we show how to write an alternative particle filter that exploits the structure of the second-order approximation to the equilibrium dynamics of an economy with stochastic volatility without the need of linear measurement errors. Second-order approximations accurately capture important implications of stochastic volatility and are convenient because they are not computationally expensive.

We proceed in two steps. First, we characterize the second-order approximation to the decision rules of a dynamic equilibrium model with stochastic volatility. Second, we demonstrate how to use this characterization to write the alternative particle filter. The key is to show how the quadratic problem associated with the evaluation of the approximated measurement density is reduced to a much simpler linear problem that only involves a matrix inversion. After we have evaluated the likelihood, we can combine it with a prior and a Markov chain Monte Carlo (McMc) algorithm to draw from the posterior distribution.

Our characterization of the second-order approximation to the decision rules is also of interest in itself. Among other things, it is useful to analyze the equilibrium of the model, to explore the shape of its impulse response functions, or to calibrate it. More concretely, we prove that:

1. The first-order approximation to the decision rules of the agents (or any other equilibrium object of interest) does not depend on volatility shocks and they are certainty equivalent.
2. The second-order approximation to the decision rules of the agents only depends on volatility shocks on terms where volatility is multiplied by the innovation to its own structural shock. For instance, if we have a productivity shock and a volatility shock to it, the only non-zero term where the volatility shock to productivity appears is the one where the volatility shock multiplies the innovation to the productivity shock. Thus, only a few of the terms in the second-order approximation are non-zero.
3. The perturbation parameter will only appear in a non-zero term where it is raised to a square. This term is a constant that corrects for precautionary behavior induced by risk.

As an application, we estimate a business cycle model of the US economy. The model incorporates stochastic volatility in the shocks that drive its dynamics and parameter drifting in the parameters that control monetary policy. In that way, we include two of the main mechanisms that researchers have highlighted to account for the time-varying volatility of US time series – heteroscedastic shocks and parameter drifting – and let the likelihood decide which of them better accounts for the data. Last, we have a model that is as rich as many of the models employed in modern quantitative macroeconomics. While estimating such a large model is a computational challenge, we wanted to demonstrate that our procedure is of practical use and to make our application a blueprint for the estimation of other dynamic equilibrium models.

Our main empirical findings are as follows. First, the posterior distribution of the parameters puts most of its mass in areas that denote a fair amount of stochastic volatility. Second, a model comparison exercise indicates that, even after controlling for stochastic volatility, the data prefer a specification where monetary policy changes over time. This finding should not be interpreted, though, as implying that volatility shocks did not play a role. It means, instead, that a successful model of the US economy requires the presence of both stochastic volatility and parameter drifting, a result that challenges the results of Sims and Zha (2006). Finally, we document the evolution of the structural shocks, of stochastic volatility, and the parameters of monetary policy. We emphasize the confluence, during the 1970s, of times of high volatility and weak responses to inflation, and, during the 1990s, of positive structural shocks and low volatility even if monetary policy was weaker than often argued. In the appendix, we construct counterfactual histories of the US data by varying some aspect of the model such as shutting down time-varying volatility or imposing alternative monetary policies.

An alternative to our stochastic volatility framework would be to work with Markov regime-switching models such as those of Bianchi (2009) or Farmer et al. (2009). These models provide a promising extra degree of flexibility in modeling aggregate dynamics. In fact, some of the fast changes in policy parameters that we document in our empirical section suggest that discrete jumps could be a good representation of the data. We hope to undertake in the future a more careful assessment of the advantages and disadvantages of stochastic volatility versus Markov regime-switching models.

Finally, even if the motivation for our approach and the application belong to macroeconomics, the tools we present are not specific to that field. One can think about the importance of estimating dynamic equilibrium models with stochastic volatility in many other fields such as finance (Bansal and Yaron, 2004) or international economics (Fernández-Villaverde et al., 2010b).

The rest of the paper is organized as follows. Section 2 introduces a generic dynamic equilibrium model with stochastic volatility to fix notation and discuss how to solve it. Section 3 explains the evaluation of the likelihood of the model. Section 4 compares our approach with continuous-time methods. Section 5 presents our application. Section 6 concludes. An extensive technical appendix includes additional material (see Appendix A).

2. Dynamic equilibrium models with stochastic volatility

2.1. The model

The set of equilibrium conditions of a wide class of dynamic equilibrium models can be written as

$$\mathbb{E}_t f\left(\mathcal{Y}_{t+1}, \mathcal{Y}_t, \mathcal{S}_{t+1}, \mathcal{S}_t, \mathcal{Z}_{t+1}, \mathcal{Z}_t; \gamma\right) = 0, \tag{1}$$

where $\mathbb{E}_t$ is the conditional expectation operator at time t, $\mathcal{Y}_t = (\mathcal{Y}_{1t}, \mathcal{Y}_{2t}, \ldots, \mathcal{Y}_{kt})'$ is the $k \times 1$ vector of observables at time t, $\mathcal{S}_t = (\mathcal{S}_{1t}, \mathcal{S}_{2t}, \ldots, \mathcal{S}_{nt})'$ is the $n \times 1$ vector of endogenous states at time t, $\mathcal{Z}_t = (\mathcal{Z}_{1t}, \mathcal{Z}_{2t}, \ldots, \mathcal{Z}_{mt})'$ is the $m \times 1$ vector of structural shocks at time t, f maps $\mathbb{R}^{2(k+n+m)}$ into $\mathbb{R}^{k+n+m}$, and γ is the $n_\gamma \times 1$ vector of parameters that describe preferences and technology. In this paper, γ is also the vector of parameters to be estimated.

We will consider models where $\mathcal{Z}_{it+1}$ follow a stochastic volatility process of the form

$$\mathcal{Z}_{it+1} = \rho_i \mathcal{Z}_{it} + \Lambda \sigma_i \sigma_{it+1} \varepsilon_{it+1} \tag{2}$$

for all $i \in \{1, \ldots, m\}$, where Λ is a perturbation parameter, σ_i is the mean volatility, and $\log \sigma_{it+1}$, the percentage deviation of the standard deviation of the innovations to the structural shocks with respect to its mean, evolves as

$$\log \sigma_{it+1} = \vartheta_i \log \sigma_{it} + \Lambda \left(1 - \vartheta_i^2\right)^{\frac{1}{2}} \eta_i u_{it+1} \tag{3}$$

for all $i \in \{1, \ldots, m\}$. The combination of levels in (2) and logs in (3) ensures a positive σ_{it+1}. We multiply the innovation in (3) by $\left(1 - \vartheta_i^2\right)^{\frac{1}{2}}$ to normalize its size by the persistence of σ_{it}. It will be clear momentarily why we specify (2) and (3) in terms of the perturbation parameter Λ. It is also convenient to write, for all $i \in \{1, \ldots, m\}$, the laws of motions for $\mathcal{Z}_{it}$ and $\log \sigma_{it}$

$$\mathcal{Z}_{it} = \rho_i \mathcal{Z}_{it-1} + \sigma_i \sigma_{it} \varepsilon_{it} \tag{4}$$

and

$$\log \sigma_{it} = \vartheta_i \log \sigma_{it-1} + \left(1 - \vartheta_i^2\right)^{\frac{1}{2}} \eta_i u_{it}. \tag{5}$$

Note that Λ appears only in Eqs. (2) and (3) but not in Eqs. (4) and (5). This is because this parameter is used to eliminate, when we later determine the point around which we approximate the equilibrium dynamics of the model, uncertainty about the future. Given the information set in Eq. (1), there is uncertainty about both $\mathcal{Z}_{it+1}$ and $\log \sigma_{it+1}$ for all $i \in \{1, \dots, m\}$, but there is no uncertainty about either $\mathcal{Z}_{it}$ or $\log \sigma_{it}$ for any $i \in \{1, \dots, m\}$.

Let $\Sigma_t = (\log \sigma_{1t}, \dots, \log \sigma_{mt})'$ be the $m \times 1$ vector of volatility shocks, $\mathcal{E}_t = (\varepsilon_{1t}, \dots, \varepsilon_{mt})'$ the $m \times 1$ vector of innovations to the structural shocks, and $\mathcal{U}_t = (u_{1t}, \dots, u_{mt})'$ the $m \times 1$ vector of innovations to the volatility shocks. We assume that $\mathcal{E}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\mathcal{U}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, where $\mathbf{0}$ is an $m \times 1$ vector of zeros and $\mathbf{I}$ is an $m \times m$ identity matrix. To ease notation, we assume that all structural shocks face volatility shocks, that the volatility shocks are uncorrelated, and that $\mathcal{E}_t$ and $\mathcal{U}_t$ are normally distributed. It is straightforward, yet cumbersome, to generalize the notation to other cases. In particular, as we will see below, to implement our particle filter, we only need to be able to simulate $\mathcal{E}_t$ and evaluate the density of $\mathcal{U}_t$.

2.2. The solution

Given Eqs. (1)–(5), the solution to the model – if one exists – that embodies equilibrium dynamics (that is, agents' optimization and market clearing conditions) is characterized by a policy function describing the evolution of the endogenous state variables

$$\mathcal{S}_{t+1} = h\left(\mathcal{S}_t, \mathcal{Z}_{t-1}, \Sigma_{t-1}, \mathcal{E}_t, \mathcal{U}_t, \Lambda; \gamma\right), \tag{6}$$

and two policy functions describing the law of motion of the observables

$$\mathcal{Y}_t = g\left(\mathcal{S}_t, \mathcal{Z}_{t-1}, \Sigma_{t-1}, \mathcal{E}_t, \mathcal{U}_t, \Lambda; \gamma\right) \tag{7}$$

and

$$\mathcal{Y}_{t+1} = g\left(\mathcal{S}_{t+1}, \mathcal{Z}_t, \Sigma_t, \Lambda\mathcal{E}_{t+1}, \Lambda\mathcal{U}_{t+1}, \Lambda; \gamma\right), \tag{8}$$

together with Eqs. (4) and (5) describing the laws of motion of the structural and volatility shocks. The policy functions h and g map $\mathbb{R}^{n+4m+1}$ into $\mathbb{R}^n$ and $\mathbb{R}^k$, respectively, and are indexed by γ.

For our purposes, it is important to define the steady state of the model. Our assumptions about the stochastic processes imply that, in the steady state, $\mathcal{Z} = \mathbf{0}$ and $\Sigma = \mathbf{0}$. Given γ and Eqs. (1)–(8), the steady state of the model is a $k \times 1$ vector of observables $\mathcal{Y} = (\mathcal{Y}_1, \mathcal{Y}_2, \dots, \mathcal{Y}_k)'$ and an $n \times 1$ vector of endogenous states $\mathcal{S} = (\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_n)'$ such that

$$\begin{aligned} &f\left(g\left(h\left(\mathcal{S}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}, 0; \gamma\right), \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}, 0; \gamma\right),\right. \\ &\quad \left. g\left(\mathcal{S}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}, 0; \gamma\right), h\left(\mathcal{S}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}, 0; \gamma\right), \mathcal{S}, \mathbf{0}, \mathbf{0}; \gamma\right) = 0. \end{aligned} \tag{9}$$

In addition, in the steady state, the following two relationships hold

$$\mathcal{S} = h\left(\mathcal{S}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}, 0; \gamma\right), \tag{10}$$

and

$$\mathcal{Y} = g\left(\mathcal{S}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}, 0; \gamma\right). \tag{11}$$

Note that, if $\Lambda = 0$, the model is in the steady state, since we eliminate any uncertainty about the future. If $\Lambda \neq 0$, the model is not in the steady state and conditions (9)–(11) do not hold. For instance, in general $\mathcal{S} \neq h\left(\mathcal{S}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \Lambda; \gamma\right)$ because of the precautionary behavior of agents.

2.3. The state-space representation

Given the solution to the model – the policy functions (6)–(8) together with Eqs. (4) and (5) – we can characterize the equilibrium dynamics of the model by its state-space representation written in terms of the transition and the observation equations.

We stack Eqs. (4)–(6) in a transition equation

$$\mathbb{S}_{t+1} = \widetilde{h}\left(\mathbb{S}_t, \Lambda; \gamma\right) + \Xi\mathbb{W}_{t+1} \tag{12}$$

that describe the evolution of the states (endogenous states, structural shocks, volatility shocks, and their innovations) as a function of lag states, the perturbation parameter, and the vector of parameters, where $\mathbb{S}_t = \left(\mathcal{S}_t', \mathcal{Z}_{t-1}', \Sigma_{t-1}', \mathcal{E}_t', \mathcal{U}_t'\right)'$ is the $(n+4m) \times 1$ vector of the states and $\widetilde{h}$ maps $\mathbb{R}^{n+4m+1}$ into $\mathbb{R}^{n+4m}$. Also, $\mathbb{W}_{t+1} = \left(\mathcal{W}_{1t+1}', \mathcal{W}_{2t+1}'\right)'$ is a $2m \times 1$ vector of random variables, $\mathcal{W}_{1t+1}$ and $\mathcal{W}_{2t+1}$ are $m \times 1$ vectors with distributions $\mathcal{W}_{1t+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\mathcal{W}_{2t+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and Ξ is a $(n+4m) \times 2m$ matrix with the top $n+2m$ rows equal to zero and the bottom $2m \times 2m$ matrix equal to the identity matrix. $\mathcal{W}_{1t+1}$ and $\mathcal{W}_{2t+1}$ share distributions with $\mathcal{E}_{t+1}$ and $\mathcal{U}_{t+1}$ respectively. If we were to change distributions for either $\mathcal{E}_{t+1}$ or $\mathcal{U}_{t+1}$ we would need to do the same with $\mathcal{W}_{1t+1}$ and $\mathcal{W}_{2t+1}$. Let us define $n_s = n + 4m$. The linearity of the second term of the right-hand side is a consequence of the autoregressive specification of the structural shocks and the evolution of their volatilities. However, through the function $\widetilde{h}$, those shocks and their volatilities can affect $\mathcal{S}_{t+1}$ non-linearly.

The measurement equation uses the policy function (7) to describe the relationship of the observables with the states, the perturbation parameter, and the vector of parameters

$$\mathcal{Y}_t = g\left(\mathbb{S}_t, \Lambda; \gamma\right). \tag{13}$$

2.4. Approximating the state-space representation

In general, when we deal with the class of dynamic equilibrium models with stochastic volatility, the policy functions h and g cannot be found explicitly. Thus, we cannot build the state-space representation described by (12) and (13). Instead, we will approximate numerically the solution of the model and use the result to generate an approximated state-space representation.

There are many different solution algorithms for dynamic equilibrium models. Among them, the perturbation method has emerged as a popular way (see Judd and Guu, 1997; Aruoba et al., 2006) to obtain higher-order Taylor series approximations to the policy functions (6)–(7) together with Eqs. (4) and (5) around the steady state. We also get the Taylor expansion of (4) because it is a non-linear function. Beyond being extremely fast for systems with a large number of state variables, perturbations are highly accurate even far away from the perturbation point (Aruoba et al., 2006), and, for cases with stochastic volatility, (Caldara et al., 2012).

The perturbation method allows the researcher to approximate the solution to the model up to any order. Since we want to analyze models with volatility shocks, we must go beyond a first-order approximation. First-order approximations are certainty equivalent, and hence, they are silent about stochastic volatility. As we will see in Section 3.1, volatility shocks only appear starting in the second-order approximation. But since we want to evaluate the likelihood function, we stop at a second-order approximation. Because of dimensionality issues, a higher-order approximation would make the evaluation of the likelihood function exceedingly challenging for current computers for models with a reasonable number of state variables.

Given a second-order approximation to the policy functions (6)–(7) together with Eqs. (4) and (5) around the steady state, the approximated state-space representation can be written in terms of two equations: the approximated transition equation and the approximated measurement equation. The approximated transition equation is

$$\widehat{\mathbb{S}}_{t+1} = \begin{pmatrix} \Psi^1_{s,1}\widehat{\mathbb{S}}_t \\ \vdots \\ \Psi^1_{s,n_s}\widehat{\mathbb{S}}_t \end{pmatrix} + \frac{1}{2}\begin{pmatrix} \widehat{\mathbb{S}}'_t\Psi^2_{s,1}\widehat{\mathbb{S}}_t \\ \vdots \\ \widehat{\mathbb{S}}'_t\Psi^2_{s,n_s}\widehat{\mathbb{S}}_t \end{pmatrix} + \frac{1}{2}\begin{pmatrix} \Psi^{\Lambda}_{s,1} \\ \vdots \\ \Psi^{\Lambda}_{s,n_s} \end{pmatrix} + \Xi\mathbb{W}_{t+1} \quad (14)$$

where $\widehat{\mathbb{S}}_t = \mathbb{S}_t - \mathbb{S}$ is the $n_s \times 1$ vector of deviations of the states from their steady-state value and $\mathbb{S} = \left(\mathcal{S}', \mathbf{0}', \mathbf{0}', \mathbf{0}', \mathbf{0}'\right)'$. The approximated measurement equation is

$$\mathcal{Y}_t - \mathcal{Y} = \begin{pmatrix} \Psi^1_{y,1}\widehat{\mathbb{S}}_t \\ \vdots \\ \Psi^1_{y,k}\widehat{\mathbb{S}}_t \end{pmatrix} + \frac{1}{2}\begin{pmatrix} \widehat{\mathbb{S}}'_t\Psi^2_{y,1}\widehat{\mathbb{S}}_t \\ \vdots \\ \widehat{\mathbb{S}}'_t\Psi^2_{y,k}\widehat{\mathbb{S}}_t \end{pmatrix} + \frac{1}{2}\begin{pmatrix} \Psi^{\Lambda}_{y,1} \\ \vdots \\ \Psi^{\Lambda}_{y,k} \end{pmatrix} \quad (15)$$

where $\mathcal{Y}$ is the steady-state value of $\mathcal{Y}_t$.

In these equations, $\Psi^1_{s,i}$ is a $1 \times n_s$ vector and $\Psi^2_{s,i}$ is an $n_s \times n_s$ matrix for $i = 1, \ldots, n_s$. The first term is the linear approximation to the transition equation for the states, while the second term is the quadratic component of the second-order approximation. Similarly, $\Psi^1_{y,i}$ is a $1 \times n_s$ vector and $\Psi^2_{y,i}$ an $n_s \times n_s$ matrix for $i = 1, \ldots, k$. The interpretation of each term is the same as before, but for the measurement equations. The term $\Psi^{\Lambda}_{s,i}$ is the constant that appears in the second-order perturbation that corrects for risk in the evolution of state $i = 1, \ldots, n_s$. Similarly, the term $\Psi^{\Lambda}_{y,i}$ is the constant correction for risk of observable $i = 1, \ldots, k$. All these vectors and matrices are non-linear functions of γ. It is important to emphasize that we are not assuming the presence of any measurement error. We will return to this point in a few paragraphs.

3. Stochastic volatility and evaluation of the likelihood

In this section, we explain how to evaluate the likelihood function of our class of dynamic equilibrium models with volatility shocks in Eq. (1) using the approximated transition and the measurement Eqs. (14) and (15). If we allow $\mathbb{Y}_t$ to be the data counterpart of our observables $\mathcal{Y}_t$, and $\mathbb{Y}^t = (\mathbb{Y}_1, \ldots, \mathbb{Y}_t)$ (with $\mathbb{Y}^0 = \{\varnothing\}$) to be their history up to time t, given γ, we can write the likelihood of $\mathbb{Y}^T$ as

$$\prod_{t=1}^{T} p\left(\mathcal{Y}_t = \mathbb{Y}_t|\mathbb{Y}^{t-1}; \gamma\right),$$

where

$$\begin{aligned} &p\left(\mathcal{Y}_t = \mathbb{Y}_t|\mathbb{Y}^{t-1}; \gamma\right) \\ &\quad = \int\int\int\int p\left(\mathcal{Y}_t = \mathbb{Y}_t|\mathcal{S}_t, \mathcal{Z}_{t-1}, \Sigma_{t-1}, \mathcal{E}_t; \gamma\right) \\ &\qquad \times p\left(\mathcal{S}_t, \mathcal{Z}_{t-1}, \Sigma_{t-1}, \mathcal{E}_t|\mathbb{Y}^{t-1}; \gamma\right) d\mathcal{S}_t d\mathcal{Z}_{t-1} d\Sigma_{t-1} d\mathcal{E}_t \end{aligned} \quad (16)$$

for all $t \in \{2, \ldots, T\}$ and

$$\begin{aligned} &p\left(\mathcal{Y}_1 = \mathbb{Y}_1; \gamma\right) \\ &\quad = \int\int\int\int p\left(\mathcal{Y}_1 = \mathbb{Y}_1|\mathcal{S}_1, \mathcal{Z}_0, \Sigma_0, \mathcal{E}_1; \gamma\right) \\ &\qquad \times p\left(\mathcal{S}_1, \mathcal{Z}_0, \Sigma_0, \mathcal{E}_1; \gamma\right) d\mathcal{S}_1 d\mathcal{Z}_0 d\Sigma_0 d\mathcal{E}_1. \end{aligned} \quad (17)$$

Note that the $\mathcal{U}_t$'s do not show up in these two expressions. It will be momentarily clear why this comes directly from our procedure below to evaluate the likelihood.

Computing this likelihood is difficult. Since we do not have analytic forms for the terms inside the integral, it cannot be evaluated exactly and deterministic numerical integration algorithms are too slow for practical use (we have four integrals per period over large dimensions). As a feasible alternative, we will show how to use a simple particle filter to obtain a simulation-based estimate of (16). Künsch (2005) proves, under weak conditions, that the particle filter delivers a consistent estimator of the likelihood function and that a central limit theorem applies. A particle filter is a sequential simulation device for filtering of non-linear and/or non-Gaussian space models (Pitt and Shephard, 1999; Doucet et al., 2001). In economics the particle filter was introduced by Fernández-Villaverde and Rubio-Ramirez (2007).

As mentioned before, the particle filter has minimal requirements: the ability to evaluate the approximated measurement density associated with the approximated measurement equation, to simulate from the approximated dynamics of the state using the approximated transition equation, and to draw from the unconditional density of the states implied by the approximated transition equation. Usually, the first requirement is the hardest. These three requirements are formally described in the following assumption.

Assumption 1. To implement the particle filter, we assume that:

1. We can evaluate the approximated measurement density
   $p\left(\mathcal{Y}_t = \mathbb{Y}_t|\mathcal{S}_t, \mathcal{Z}_{t-1}, \Sigma_{t-1}, \mathcal{E}_t; \gamma\right).$
2. We can simulate from the approximated transition equation
   $\left(\mathcal{S}'_{t+1}, \mathcal{Z}'_t, \Sigma'_t, \mathcal{E}'_{t+1}\right)' | \left(\mathcal{S}'_t, \mathcal{Z}'_{t-1}, \Sigma'_{t-1}, \mathcal{E}'_t\right)', \mathcal{F}\left(\mathbb{Y}^t\right); \gamma$
   for all $t \in \{1, \ldots, T\}$, where $\mathcal{F}\left(\mathbb{Y}^t\right)$ is the filtration of $\mathbb{Y}^t$.
3. We can draw from the unconditional distribution implied by the approximated transition equation
   $p\left(\mathcal{S}_t, \mathcal{Z}_{t-1}, \Sigma_{t-1}, \mathcal{E}_t; \gamma\right).$

The second requirement asks for the filtration of $\mathbb{Y}^t$, $\mathcal{F}\left(\mathbb{Y}^t\right)$, because we will need to evaluate the volatility shocks. The last requirement can be easily implemented using the results in Santos and Peralta-Alva (2005). Given our assumption about $\mathcal{E}_{t+1}$ and the quadratic form of the approximated transition equation (14), the second requirement easily holds. A key novelty of this paper is that we show how the class of dynamic equilibrium models with volatility shocks considered here also satisfies the first requirement.

For our class of models, conditional on having N draws of $\left\{s^i_t, z^i_{t-1}, \sigma^i_{t-1}, \varepsilon^i_t\right\}_{i=1}^N$ from the density $p(\mathcal{S}_t, \mathcal{Z}_{t-1}, \Sigma_{t-1}, \mathcal{E}_t|\mathbb{Y}^{t-1}; \gamma)$, each integral (16) can be consistently approximated by

$$\begin{aligned} &p\left(\mathcal{Y}_t = \mathbb{Y}_t|\mathbb{Y}^{t-1}; \gamma\right) \\ &\quad \simeq \frac{1}{N}\sum_{i=1}^{N} p\left(\mathcal{Y}_t = \mathbb{Y}_t|s^i_t, z^i_{t-1}, \sigma^i_{t-1}, \varepsilon^i_t; \gamma\right) \end{aligned} \quad (18)$$

for all $t \in \{2, \ldots, T\}$. For $t = 1$, we need N draws from the density $p\left(\mathcal{S}_t, \mathcal{Z}_{t-1}, \Sigma_{t-1}, \mathcal{E}_t; \gamma\right)$, so that the integral (17) can be consistently approximated by

$$p\left(\mathcal{Y}_1 = \mathbb{Y}_1; \gamma\right) \simeq \frac{1}{N}\sum_{i=1}^{N} p\left(\mathcal{Y}_1 = \mathbb{Y}_1|s^i_1, z^i_0, \sigma^i_0, \varepsilon^i_1; \gamma\right). \quad (19)$$

We know that we can make the draws because requirement 3 in Assumption 1 holds.

In our framework, checking whether requirement 1 in Assumption 1 holds means checking whether we can evaluate, for each draw in $\left\{s^i_t, z^i_{t-1}, \sigma^i_{t-1}, \varepsilon^i_t\right\}_{i=1}^N$, the approximated measurement density

$$p\left(\mathcal{Y}_t = \mathbb{Y}_t|s^i_t, z^i_{t-1}, \sigma^i_{t-1}, \varepsilon^i_t; \gamma\right) \quad (20)$$

for all $t \in \{1, \ldots, T\}$. This evaluation step is crucial, not only because it is required to compute (18) and (19) but also because, as we will explain momentarily, our particle filter needs to evaluate

(20) to resample from $p\left(\mathcal{S}_t, \mathcal{Z}_{t-1}, \Sigma_{t-1}, \mathcal{E}_t | \mathbb{Y}^{t-1}; \gamma\right)$ and get draws from $p\left(\mathcal{S}_{t+1}, \mathcal{Z}_t, \Sigma_t, \mathcal{E}_{t+1} | \mathbb{Y}^t; \gamma\right)$ to obtain $\left\{s_t^i, z_{t-1}^i, \sigma_{t-1}^i, \varepsilon_t^i\right\}_{i=1}^N$ for all $t \in \{2, \ldots, T\}$ recursively.

To check how we can evaluate the approximated measurement density (20), we rewrite (15) in terms of draws $\left(s_t^{i\prime}, z_{t-1}^{i\prime}, \sigma_{t-1}^{i\prime}, \varepsilon_t^{i\prime}\right)'$ and $\mathbb{Y}_t$, instead of $\left(\mathcal{S}_t', \mathcal{Z}_{t-1}', \Sigma_{t-1}', \mathcal{E}_t'\right)'$ and $\mathcal{Y}_t$. Thus, the approximated measurement equation (15) becomes

$$\mathbb{Y}_t - \mathcal{Y} = \begin{pmatrix} \Psi_{y,1}^1 \widehat{\mathbb{S}}_t^i \\ \vdots \\ \Psi_{y,k}^1 \widehat{\mathbb{S}}_t^i \end{pmatrix} + \frac{1}{2} \begin{pmatrix} \widehat{\mathbb{S}}_t^{i\prime} \Psi_{y,1}^2 \widehat{\mathbb{S}}_t^i \\ \vdots \\ \widehat{\mathbb{S}}_t^{i\prime} \Psi_{y,k}^2 \widehat{\mathbb{S}}_t^i \end{pmatrix} + \frac{1}{2} \begin{pmatrix} \Psi_{y,1}^{\Lambda} \\ \vdots \\ \Psi_{y,k}^{\Lambda} \end{pmatrix} \tag{21}$$

where $\widehat{\mathbb{S}}_t^i = \mathbb{S}_t^i - \mathbb{S}$ and $\mathbb{S}_t^i = \left(s_t^{i\prime}, z_{t-1}^{i\prime}, \sigma_{t-1}^{i\prime}, \varepsilon_t^{i\prime}, \mathcal{U}_t'\right)'$. The new approximated measurement equation (21) implies that evaluating the approximated measurement density (20) involves solving the system of quadratic equations

$$\mathbb{Y}_t - \mathcal{Y} - \begin{pmatrix} \Psi_{y,1}^1 \widehat{\mathbb{S}}_t^i \\ \vdots \\ \Psi_{y,k}^1 \widehat{\mathbb{S}}_t^i \end{pmatrix} - \frac{1}{2} \begin{pmatrix} \widehat{\mathbb{S}}_t^{i\prime} \Psi_{y,1}^2 \widehat{\mathbb{S}}_t^i \\ \vdots \\ \widehat{\mathbb{S}}_t^{i\prime} \Psi_{y,k}^2 \widehat{\mathbb{S}}_t^i \end{pmatrix} - \frac{1}{2} \begin{pmatrix} \Psi_{y,1}^{\Lambda} \\ \vdots \\ \Psi_{y,k}^{\Lambda} \end{pmatrix} = 0 \tag{22}$$

for $\mathcal{U}_t$ given $\mathbb{Y}_t, s_t^i, z_{t-1}^i, \sigma_{t-1}^i$, and ε_t^i.

Solving this system is non-trivial. Since the system is quadratic, we may have either no solution or several different ones. In fact, it is even hard to know how many solutions there are. But even if we knew the number of solutions, we are not aware of any accurate and efficient method to solve quadratic problems that finds all the solutions. This difficulty seemingly prevents us from achieving our goal of evaluating the likelihood function.

A solution would be to introduce, as Fernández-Villaverde and Rubio-Ramirez (2007) did, a $k \times 1$ vector of linear measurement errors and solve for those instead of $\mathcal{U}_t$. In this case the system would have a unique, easy-to-find solution. However, there are three reasons, in increasing order of importance, why this solution is not satisfactory:

1. Although measurement errors are plausible, their presence complicates the interpretation of any empirical results. In particular, we are interested in measuring how heteroscedastic structural shocks help in accounting for the data and measurement errors can confound us.
2. The absence of measurement errors will help us to illustrate below how dynamic equilibrium models with volatility shocks have a profusion of shocks that we can exploit in our estimation.
3. As we will also show below, volatility shocks would enter linearly (conditional on the draw) in the system equations. Since, by definition, linear measurement errors would also enter in the same fashion, it would be hard to identify one apart from the others.

Our alternative in this paper is to realize that considering stochastic volatility converts the above-described quadratic system into a linear one. Hence, if a rank condition is satisfied, the system (22) has only one solution and the solution can be found by inverting a matrix. Thus, requirement 1 in Assumption 1 holds and we can use the particle filter. The core of the argument is to note that, when volatility shocks are considered, the policy functions share a peculiar pattern that we can exploit.

3.1. Structure of the solution

Our first step will characterize the first- and second-order derivatives of the policy functions h and g evaluated at the steady state. Then, we will describe an interesting pattern in these derivatives. The second step will take advantage of the pattern to show that, when the number of volatility shocks equals the number of observables, our quadratic system becomes a linear one. This characterization is important both for estimation and, more generally, for the analysis of perturbation solutions to dynamic equilibrium models with stochastic volatility.

3.1.1. First- and second-order derivatives of the policy functions

Let us begin with the characterization of the first- and second-order derivatives of the policy functions. The following theorem shows that the first-order derivatives of h and g with respect to any component of $\mathcal{U}_t$ and Σ_{t-1} evaluated at the steady state are zero; that is, volatility shocks and their innovations do not affect the linear component of the optimal decision rule of the agents. The same occurs with the perturbation parameter Λ. A similar result has been established by Schmitt-Grohé and Uribe (2004) for the homoscedastic shocks case.

Theorem 2. *Let us denote $[\Upsilon_\omega]_j^i$ as the derivative of the ith element of generic function Υ with respect to the jth element of generic variable ω evaluated at the non-stochastic steady state (where we drop this index if ω is unidimensional). Then, for the dynamic equilibrium model specified in Eq. (1), we have that $\left[h_{\Sigma_{t-1}}\right]_j^{i_1} = \left[g_{\Sigma_{t-1}}\right]_j^{i_2} = \left[h_{\mathcal{U}_t}\right]_j^{i_1} = \left[g_{\mathcal{U}_t}\right]_j^{i_2} = [h_\Lambda]^{i_1} = [g_\Lambda]^{i_2} = 0$ for $i_1 \in \{1, \ldots, n\}$, $i_2 \in \{1, \ldots, k\}$, and $j \in \{1, \ldots, m\}$.*

Proof. See appendix 2.1. ∎

The second theorem shows, among other things, that the second partial derivatives of h and g with respect to either $\log \sigma_{it-1}$ or $u_{i,t}$ and any other variable but $\varepsilon_{i,t}$ are also zero for any $i \in \{1, \ldots, m\}$.

Theorem 3. *Furthermore, if we denote $\left[\Upsilon_{\omega\xi}\right]_{j_1,j_2}^i$ as the derivative of the ith element of generic function Υ with respect to the j_1th element of generic variable ω and the j_2th element of generic variable ξ evaluated at the non-stochastic steady state (where again we drop the index for unidimensional variables), we have that $\left[h_{\Lambda,\mathcal{S}_t}\right]_j^{i_1} = \left[g_{\Lambda,\mathcal{S}_t}\right]_j^{i_2} = 0$ for $i_1 \in \{1, \ldots, n\}$, $i_2 \in \{1, \ldots, k\}$, and $j \in \{1, \ldots, n\}$,*

$$\left[h_{\Lambda,\mathcal{Z}_{t-1}}\right]_j^{i_1} = \left[g_{\Lambda,\mathcal{Z}_{t-1}}\right]_j^{i_2} = \left[h_{\Lambda,\Sigma_{t-1}}\right]_j^{i_1} = \left[g_{\Lambda,\Sigma_{t-1}}\right]_j^{i_2}$$
$$= \left[h_{\Lambda,\mathcal{E}_t}\right]_j^{i_1} = \left[g_{\Lambda,\mathcal{E}_t}\right]_j^{i_2} = \left[h_{\Lambda,\mathcal{U}_t}\right]_j^{i_1} = \left[g_{\Lambda,\mathcal{U}_t}\right]_j^{i_2} = 0$$

for $i_1 \in \{1, \ldots, n\}$, $i_2 \in \{1, \ldots, k\}$, and $j \in \{1, \ldots, m\}$,

$$\left[h_{\mathcal{S}_t,\Sigma_{t-1}}\right]_{j_1,j_2}^{i_1} = \left[g_{\mathcal{S}_t,\Sigma_{t-1}}\right]_{j_1,j_2}^{i_2} = \left[h_{\mathcal{S}_t,\mathcal{U}_t}\right]_{j_1,j_2}^{i_1} = \left[g_{\mathcal{S}_t,\mathcal{U}_t}\right]_{j_1,j_2}^{i_2} = 0$$

for $i_1 \in \{1, \ldots, n\}$, $i_2 \in \{1, \ldots, k\}$, $j_1 \in \{1, \ldots, n\}$, and $j_2 \in \{1, \ldots, m\}$,

$$\left[h_{\mathcal{Z}_{t-1},\Sigma_{t-1}}\right]_{j_1,j_2}^{i_1} = \left[g_{\mathcal{Z}_{t-1},\Sigma_{t-1}}\right]_{j_1,j_2}^{i_2} = \left[h_{\Sigma_{t-1},\Sigma_{t-1}}\right]_{j_1,j_2}^{i_1}$$
$$= \left[g_{\Sigma_{t-1},\Sigma_{t-1}}\right]_{j_1,j_2}^{i_2} = 0$$

and

$$\left[h_{\mathcal{Z}_{t-1},\mathcal{U}_t}\right]_{j_1,j_2}^{i_1} = \left[g_{\mathcal{Z}_{t-1},\mathcal{U}_t}\right]_{j_1,j_2}^{i_2} = \left[h_{\Sigma_{t-1},\mathcal{U}_t}\right]_{j_1,j_2}^{i_1} = \left[g_{\Sigma_{t-1},\mathcal{U}_t}\right]_{j_1,j_2}^{i_2}$$
$$= \left[h_{\mathcal{U}_t,\mathcal{U}_t}\right]_{j_1,j_2}^{i_1} = \left[g_{\mathcal{U}_t,\mathcal{U}_t}\right]_{j_1,j_2}^{i_2} = 0$$

for $i_1 \in \{1, \ldots, n\}$, $i_2 \in \{1, \ldots, k\}$, and $j_1, j_2 \in \{1, \ldots, m\}$, and

$$\left[h_{\mathcal{E}_t,\Sigma_{t-1}}\right]_{j_1,j_2}^{i_1} = \left[g_{\mathcal{E}_t,\Sigma_{t-1}}\right]_{j_1,j_2}^{i_2} = \left[h_{\mathcal{E}_t,\mathcal{U}_t}\right]_{j_1,j_2}^{i_1} = \left[g_{\mathcal{E}_t,\mathcal{U}_t}\right]_{j_1,j_2}^{i_2} = 0$$

for $i_1 \in \{1, \ldots, n\}$, $i_2 \in \{1, \ldots, k\}$, and $j_1, j_2 \in \{1, \ldots, m\}$ if $j_1 \neq j_2$.

Proof. See appendix 2.2. ∎

Table 3.1
Second derivatives.

$\mathcal{S}_t\mathcal{S}_t \neq 0$	$\mathcal{S}_t\mathcal{Z}_{t-1} \neq 0$	$\mathcal{S}_t\Sigma_{t-1} = 0$	$\mathcal{S}_t\mathcal{E}_t \neq 0$	$\mathcal{S}_t\mathcal{U}_t = 0$	$\mathcal{S}_t\Lambda = 0$
	$\mathcal{Z}_{t-1}\mathcal{Z}_{t-1} \neq 0$	$\mathcal{Z}_{t-1}\Sigma_{t-1} = 0$	$\mathcal{Z}_{t-1}\mathcal{E}_t \neq 0$	$\mathcal{Z}_{t-1}\mathcal{U}_t = 0$	$\mathcal{Z}_{t-1}\Lambda = 0$
		$\Sigma_{t-1}\Sigma_{t-1} = 0$	$\Sigma_{t-1}\mathcal{E}_t \neq 0^*$	$\Sigma_{t-1}\mathcal{U}_t = 0$	$\Sigma_{t-1}\Lambda = 0$
			$\mathcal{E}_t\mathcal{E}_t \neq 0$	$\mathcal{E}_t\mathcal{U}_t \neq 0^*$	$\mathcal{E}_t\Lambda = 0$
				$\mathcal{U}_t\mathcal{U}_t = 0$	$\mathcal{U}_t\Lambda = 0$
					$\Lambda\Lambda \neq 0$

We clarify the statement of Theorem 3 with Table 3.1, in which we characterize the second derivatives of the functions h and g with respect to the different variables ($\mathcal{S}_t$, $\mathcal{Z}_{t-1}$, Σ_{t-1}, $\mathcal{E}_t$, $\mathcal{U}_t$, Λ) (for ease of exposition, in Table 3.1 we are not being explicit about the dimensions of the matrices: it is a qualitative description of the relevant derivatives.) This pattern is both interesting and useful. The way to read Table 3.1 is as follows. Take an arbitrary entry, for instance, entry (1,2), $\mathcal{S}_t\mathcal{Z}_{t-1} \neq 0$. In this entry, we state that the cross-derivatives of h and g with respect to $\mathcal{S}_t$ and $\mathcal{Z}_{t-1}$ are different from zero (the table is upper triangular because, given the symmetry of second derivatives, we do not need to report those entries). Similarly, entry (3,5), $\Sigma_{t-1}\mathcal{U}_t = 0$ tells us that the cross-derivatives of h and g with respect to Σ_{t-1} and $\mathcal{U}_t$ are all zero. Entries (3,4) and (4,5) have a "*" to denote that the only cross-derivatives of those entries that are different from zero are those that correspond to the same index j (that is, the cross-derivatives of each innovation to the structural shocks with respect to its own volatility shock and the cross-derivatives of the innovation to the structural shocks to the innovation to its own volatility shock). The lower triangular part of the table is empty because of the symmetry of second derivatives.

Table 3.1 tells us that, of the 21 possible sets of second derivatives, 12 are zero and 9 are not. The implications for the decision rules of agents and for the equilibrium function are striking. The perturbation parameter, Λ, will only have a coefficient different from zero in the term where it appears in a square by itself. This term is a constant that corrects for precautionary behavior induced by risk. Volatility shocks, Σ_{t-1}, appear with coefficients different from zero only in the term $\Sigma_{t-1}\mathcal{E}_t$ where they are multiplied by the innovation to its own structural shock $\mathcal{E}_t$. Finally, innovations to the volatility shocks, $\mathcal{U}_t$, also appear with coefficients different from zero when they show up with the innovation to their own structural shock $\mathcal{E}_t$. Hence, of the terms that complicate the evaluation of the approximated measurement density, only the ones associated with $\left[h_{\mathcal{E}_t,\mathcal{U}_t}\right]^{i_1}_{j_1,j_1}$ and $\left[g_{\mathcal{E}_t,\mathcal{U}_t}\right]^{i_2}_{j_1,j_1}$ are non-zero.

3.1.2. Evaluating the likelihood using the particle filter

The second step is to use Theorems 2 and 3 to show that the system (22) is linear and has only one solution. Corollary 4 shows that the pattern described in Table 3.1 has an important implication for the structure of the approximated measurement equation (21).

Corollary 4. *The approximated measurement equation (21) can be written as*

$$\mathbb{Y}_t - \mathcal{Y} = \begin{pmatrix} \widetilde{\Psi}^1_{y,1}\widetilde{\widetilde{\mathbb{S}}}^i_t \\ \vdots \\ \widetilde{\Psi}^1_{y,k}\widetilde{\widetilde{\mathbb{S}}}^i_t \end{pmatrix} + \frac{1}{2}\begin{pmatrix} \widetilde{\widetilde{\mathbb{S}}}^{i\prime}_t\widetilde{\Psi}^{2,1}_{y,1}\widetilde{\widetilde{\mathbb{S}}}^i_t \\ \vdots \\ \widetilde{\widetilde{\mathbb{S}}}^{i\prime}_t\widetilde{\Psi}^{2,1}_{y,k}\widetilde{\widetilde{\mathbb{S}}}^i_t \end{pmatrix} + \frac{1}{2}\begin{pmatrix} \Psi^{\Lambda}_{y,1} \\ \vdots \\ \Psi^{\Lambda}_{y,k} \end{pmatrix}$$
$$+ \begin{pmatrix} \varepsilon^{i\prime}_t\widetilde{\Psi}^{2,2}_{y,1} \\ \vdots \\ \varepsilon^{i\prime}_t\widetilde{\Psi}^{2,2}_{y,k} \end{pmatrix}\sigma^i_{t-1} + \begin{pmatrix} \varepsilon^{i\prime}_t\widetilde{\Psi}^{2,3}_{y,1} \\ \vdots \\ \varepsilon^{i\prime}_t\widetilde{\Psi}^{2,3}_{y,k} \end{pmatrix}\mathcal{U}_t$$

where $\widetilde{\mathbb{S}}^i_t = \left(s^{i\prime}_t, z^{i\prime}_{t-1}, \varepsilon^{i\prime}_t\right)'$ is the $(n+2m) \times 1$ vector of the simulated states without the stochastic volatility components (that is, endogenous states, structural shocks, and their innovations), $\widetilde{\mathbb{S}} = \left(\mathcal{S}', \mathbf{0}', \mathbf{0}'\right)'$ is its steady state, and $\widetilde{\widetilde{\mathbb{S}}}^i_t = \widetilde{\mathbb{S}}^i_t - \widetilde{\mathbb{S}}$ is its deviation from the steady state. Let us define $\widetilde{n}_s = n + 2m$. The matrix $\widetilde{\Psi}^1_{y,j}$ is a $1 \times \widetilde{n}_s$ vector that represents the first-order component of the second-order approximation to the law of motion for the measurement equation as a function of $\widetilde{\widetilde{\mathbb{S}}}_t$, where $\widetilde{\mathbb{S}}_t = \left(\mathcal{S}'_t, \mathcal{Z}'_{t-1}, \mathcal{E}'_t\right)'$ and $\widetilde{\widetilde{\mathbb{S}}}_t = \widetilde{\mathbb{S}}_t - \widetilde{\mathbb{S}}$, for $j = 1, \ldots, k$. The matrix $\widetilde{\Psi}^{2,1}_{y,j}$ is an $\widetilde{n}_s \times \widetilde{n}_s$ matrix that represents the second-order component of the second-order approximation to the law of motion for the measurement equation as a function of $\widetilde{\widetilde{\mathbb{S}}}_t$ for $j = 1, \ldots, k$. The matrix $\widetilde{\Psi}^{2,2}_{y,j}$ is an $m \times m$ matrix that represents the second-order component of the second-order approximation to the law of motion for the measurement equation as a function of $\mathcal{E}_t$ and Σ_{t-1} for $j = 1, \ldots, k$. The matrix $\widetilde{\Psi}^{2,3}_{y,j}$ is an $m \times m$ matrix that represents the second-order term of the approximated law of motion for the measurement equation as a function of $\mathcal{E}_t$ and $\mathcal{U}_t$ for $j = 1, \ldots, k$.

We are now ready to show that the system (22) is linear, and if a rank condition is satisfied, it has only one solution.

Define

$$\mathbb{A}\left(\mathbb{Y}_t, s^i_t, z^i_{t-1}, \sigma^i_{t-1}, \varepsilon^i_t; \gamma\right)$$
$$\equiv \mathbb{Y}_t - \mathcal{Y} - \begin{pmatrix} \widetilde{\Psi}^1_{y,1}\widetilde{\widetilde{\mathbb{S}}}^i_t \\ \vdots \\ \widetilde{\Psi}^1_{y,k}\widetilde{\widetilde{\mathbb{S}}}^i_t \end{pmatrix} - \frac{1}{2}\begin{pmatrix} \widetilde{\widetilde{\mathbb{S}}}^{i\prime}_t\widetilde{\Psi}^{2,1}_{y,1}\widetilde{\widetilde{\mathbb{S}}}^i_t \\ \vdots \\ \widetilde{\widetilde{\mathbb{S}}}^{i\prime}_t\widetilde{\Psi}^{2,1}_{y,k}\widetilde{\widetilde{\mathbb{S}}}^i_t \end{pmatrix}$$
$$- \frac{1}{2}\begin{pmatrix} \Psi^{\Lambda}_{y,1} \\ \vdots \\ \Psi^{\Lambda}_{y,k} \end{pmatrix} - \begin{pmatrix} \varepsilon^{i\prime}_t\widetilde{\Psi}^{2,2}_{y,1} \\ \vdots \\ \varepsilon^{i\prime}_t\widetilde{\Psi}^{2,2}_{y,k} \end{pmatrix}\sigma^i_{t-1}$$

and

$$\mathbb{B}\left(\varepsilon^i_t; \gamma\right) \equiv \left(\left(\varepsilon^{i\prime}_t\widetilde{\Psi}^{2,3}_{y,1}\right)' \quad \ldots \quad \left(\varepsilon^{i\prime}_t\widetilde{\Psi}^{2,3}_{y,k}\right)'\right)'.$$

Let $k = m$, then $\mathbb{B}\left(\varepsilon^i_t; \gamma\right)$ is a $k \times k$ matrix. If $\mathbb{B}\left(\varepsilon^i_t; \gamma\right)$ is full rank, the solution to the system (22) can be written as

$$\mathcal{U}_t = \mathbb{B}^{-1}\left(\varepsilon^i_t; \gamma\right)\mathbb{A}\left(\mathbb{Y}_t, s^i_t, z^i_{t-1}, \sigma^i_{t-1}, \varepsilon^i_t; \gamma\right).$$

The next theorem shows how to use this solution to evaluate the approximated measurement density.

Theorem 5. *Let $k = m$, then $\mathbb{B}\left(\varepsilon^i_t; \gamma\right)$ is a $k \times k$ matrix. If $\mathbb{B}\left(\varepsilon^i_t; \gamma\right)$ is full rank, the approximated measurement density can be written as*

$$p\left(\mathcal{Y}_t = \mathbb{Y}_t | s^i_t, z^i_{t-1}, \sigma^i_{t-1}, \varepsilon^i_t; \gamma\right)$$
$$= \det \mathbb{B}^{-1}\left(\varepsilon^i_t; \gamma\right) p\left(\mathcal{U}_t = \mathbb{B}^{-1}\left(\varepsilon^i_t; \gamma\right)\right.$$
$$\left.\times\ \mathbb{A}\left(\mathbb{Y}_t, s^i_t, z^i_{t-1}, \sigma^i_{t-1}, \varepsilon^i_t; \gamma\right)\right) \tag{23}$$

for each draw in $\left\{s^i_t, z^i_{t-1}, \sigma^i_{t-1}, \varepsilon^i_t\right\}^N_{i=1}$ for all $t \in \{1, \ldots, T\}$, which can be evaluated given that we know $\mathbb{B}^{-1}\left(\varepsilon^i_t; \gamma\right)$, $\mathbb{A}\left(\mathbb{Y}_t, s^i_t, z^i_{t-1}, \sigma^i_{t-1}, \varepsilon^i_t; \gamma\right)$, and the distribution of $\mathcal{U}_t$.

Proof. The theorem is a straightforward application of the change of variables theorem. ■

Theorem 5 shows that requirement 1 in Assumption 1 holds and, thus, we can apply our particle filter. We are also requiring $B\left(\varepsilon_t^i;\gamma\right)$ to be full rank. When can $B\left(\varepsilon_t^i;\gamma\right)$ not be full rank? $B\left(\varepsilon_t^i;\gamma\right)$ would fail to have full rank when the impact of volatility innovations is identical across several elements of $\mathbb{Y}_t$. This would mean that $\mathcal{Y}_t$ lacks enough information to tell volatility shocks apart and that, to estimate the model, we need a new set of observables.

Note that Theorem 5 assumes $k = m$. Given the notation in Section 2.1, this means that the number of structural shocks equals the number of observables. This is not always necessary. What the theorem needs is that the number of volatility shocks equals the number of observables. Since, to simplify notation, we have assumed that all structural shocks face volatility shocks, the number of structural shocks equals the number of observables. As mentioned in Section 2.1, we could have structural shocks that do not face volatility shocks (this will be the case in our application to follow). In that case, we could have more structural shocks than observables. From the theorem, it is also clear that if we used $\mathcal{E}_t$ rather than $\mathcal{U}_t$ to compute the approximated measurement equation, we would have to solve a quadratic system, a challenging task.

An outline of the algorithm in pseudo-code is:

Step 0: `Set` $t \rightsquigarrow 1$. `Sample` N `values` $\left\{s_t^i, z_{t-1}^i, \sigma_{t-1}^i, \varepsilon_t^i\right\}_{i=1}^N$ `from` $p\left(\mathcal{S}_t, \mathcal{Z}_{t-1}, \Sigma_{t-1}, \mathcal{E}_t;\gamma\right)$.

Step 1: `Compute`

$$p\left(\mathcal{Y}_t = \mathbb{Y}_t|\mathbb{Y}^{t-1};\gamma\right) \simeq \frac{1}{N}\sum_{i=1}^{N}\det \mathbb{B}^{-1}\left(\varepsilon_t^i;\gamma\right) p(\mathcal{U}_t = \mathbb{B}^{-1}\left(\varepsilon_t^i;\gamma\right)\mathbb{A}(\mathbb{Y}_t, s_t^i, z_{t-1}^i, \sigma_{t-1}^i, \varepsilon_t^i;\gamma))$$

`using expression` (23) `and the importance weights for each draw`

$$q_t^i = \frac{\det \mathbb{B}^{-1}\left(\varepsilon_t^i;\gamma\right) p\left(\mathcal{U}_t = \mathbb{B}^{-1}\left(\varepsilon_t^i;\gamma\right)\mathbb{A}\left(\mathbb{Y}_t, s_t^i, z_{t-1}^i, \sigma_{t-1}^i, \varepsilon_t^i;\gamma\right)\right)}{\sum_{i=1}^{N}\det \mathbb{B}^{-1}\left(\varepsilon_t^i;\gamma\right) p\left(\mathcal{U}_t = \mathbb{B}^{-1}\left(\varepsilon_t^i;\gamma\right)\mathbb{A}\left(\mathbb{Y}_t, s_t^i, z_{t-1}^i, \sigma_{t-1}^i, \varepsilon_t^i;\gamma\right)\right)}.$$

Step 2: `Sample` N `times with replacement from` $\left\{s_{t|t-1}^i, z_{t-1|t-1}^i, \sigma_{t-1|t-1}^i, \varepsilon_{t|t-1}^i\right\}_{i=1}^N$ `and probability` $\left\{q_t^i\right\}_{i=1}^N$. `This delivers` $\left\{s_{t|t}^i, z_{t-1|t}^i, \sigma_{t-1|t}^i, \varepsilon_{t|t}^i\right\}_{i=1}^N$.

Step 3: `Simulate` $\left\{s_{t+1}^i, z_t^i, \sigma_t^i, \varepsilon_{t+1}^i\right\}_{i=1}^N$ `from the approximated transition equation` $\left(\mathcal{S}_{t+1}', \mathcal{Z}_t', \Sigma_t', \mathcal{E}_{t+1}'\right)' \mid \left(s_{t|t}^{i\prime}, z_{t-1|t}^{i\prime}, \sigma_{t-1|t}^{i\prime}, \varepsilon_{t|t}^{i\prime}\right)', \mathcal{F}\left(\mathbb{Y}^t\right);\gamma$.

Step 4: `If` $t < T$, `set` $t \rightsquigarrow t+1$ `and go to step 1. Otherwise stop.`

Once we have evaluated the likelihood, we can nest this algorithm either with an McMc to perform Bayesian inference (as done in our application; see Flury and Shephard, 2011, for technical details) or with some optimization algorithm to undertake maximum likelihood estimation (as done, in a model without volatility shocks, by Binsbergen et al. (2012)). In this last case, care must be taken to use an optimization algorithm that does not rely on derivatives, as the particle filter implies an evaluation of the likelihood function that is not differentiable.

4. Comparison with continuous-time models

As is common in the business-cycle literature, we wrote our generic dynamic equilibrium model in discrete time. However, much research in finance and, increasingly, in macroeconomics is using continuous-time models. Thus, it is useful to sketch how we could adapt our framework to continuous time:

1. We would write the equilibrium conditions of the continuous-time model as in Eq. (1). The main difference with discrete time is that, often, it is not possible to eliminate from those conditions the value functions (and their partial derivatives) of the agents of the model. This is not particularly problematic beyond increasing the number of equilibrium conditions (as we will need to have the concentrated Hamilton–Bellman–Jacobi equations defining those value functions in a recursive way).
2. We would solve for the policy functions of the agents using the continuous-time version of the perturbation method outlined in Section 2.4 of this paper. Judd (1998, chapters 13 and 14) discusses how to apply perturbation methods to continuous-time dynamic equilibrium models. Parra-Álvarez (2013) applies the method to the solution of business cycle models. The solution of the model would give us a multivariate diffusion process for the evolution of the states (the equivalent of our transition equation (12)) and a density for the observables (the equivalent of our transition equation (7)). An alternative would be solving the Hamilton–Bellman–Jacobi equation of the agents using a projection method as in Fernández-Villaverde et al. (2013).
3. We would then use the diffusion process and the density to build a continuous-time state-space representation analogous to Eqs. (14) and (15). This is done, for example, in Chib et al. (2004) for several finance models.
4. We would then use the methods proposed by Aït-Sahalia (2002, 2008), and Aït-Sahalia and Kimmel (2007) to find closed-form expansions for the log-likelihood function of the model. As documented in those papers, since the coefficients of the expansion are calculated explicitly by exploiting the special structure afforded by the diffusion model, the computations are fast and efficient. Then, we can either maximize the log-likelihood or nest it inside an McMc algorithm (Stramer et al., 2010).

While this approach is promising, it has not been applied in macroeconomics and it remains to assess its performance in real-life applications. In comparison, the discrete-time approach to the solution and estimation of dynamic equilibrium models has been tested in dozens of empirical applications. Therefore (and in addition to the fact that discrete time is still more popular in macroeconomics), we prefer to explore how to handle dynamic equilibrium models with stochastic volatility first in a discrete-time framework and leave the analysis of the continuous-time case for ongoing research.

5. An application: a business cycle model with stochastic volatility

As an illustration of our procedure, we estimate a business cycle model of the US economy with nominal rigidities and stochastic volatility. We will show: (1) how we can characterize posterior distributions of the parameters of interest and (2) how we can recover and analyze the smoothed structural and volatility shocks. Those are two of the most relevant exercises in terms of the estimation of dynamic equilibrium models. Once the estimation has been undertaken, there are many potential exercises. For instance, in the appendix, we include several of them such as: (1) finding the impulse response functions (IRFs) of the model; (2) evaluating the fit of the model in comparison with some alternatives; and (3) building counterfactuals and running alternative histories of the evolution of the US economy.

The model we present is a natural example for this paper because it is the base of much applied policy analysis. We will depart only along two dimensions from the standard specification: we will have stochastic volatility in the shocks that drive the dynamics of the economy and parameter drifting in the monetary policy rule. In that way, the likelihood has the chance of picking between two of the alternatives that the literature has highlighted to account for the well-documented time-varying volatility of the US aggregate time series, either a reduced volatility of the shocks that hit the economy (Sims and Zha, 2006) or a different monetary policy (Clarida et al., 2000), which makes the application of interest in itself.

5.1. Households

The economy is populated by a continuum of households indexed by j and preferences:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t d_t \left\{ \log\left(c_{jt} - hc_{jt-1}\right) + \upsilon \log\left(\frac{m_{jt}}{p_t}\right) - \varphi_t \psi \frac{l_{jt}^{1+\vartheta}}{1+\vartheta} \right\},$$

which are separable in consumption, c_{jt}, real money balances, m_{jt}/p_t, and hours worked, l_{jt}. In our notation, $\mathbb{E}_0$ is the conditional expectations operator, β is the discount factor, h controls habit persistence, ϑ is the inverse of the Frisch labor supply elasticity, d_t is a shifter to intertemporal preference that follows $\log d_t = \rho_d \log d_{t-1} + \sigma_d \sigma_{dt} \varepsilon_{dt}$ where $\varepsilon_{dt} \sim \mathcal{N}(0, 1)$, and φ_t is a labor supply shifter that evolves as $\log \varphi_t = \rho_\varphi \log \varphi_{t-1} + \sigma_\varphi \sigma_{\varphi t} \varepsilon_{\varphi t}$ where $\varepsilon_{\varphi t} \sim \mathcal{N}(0, 1)$. The two preference shocks are common to all households.

The principal novelty of these preferences is that, for both shifters d_t and φ_t, the standard deviations, σ_{dt} and $\sigma_{\varphi t}$, of their innovations, ε_{dt} and $\varepsilon_{\varphi t}$, are indexed by time; that is, they stochastically move according to:

$$\log \sigma_{dt} = \rho_{\sigma_d} \log \sigma_{dt-1} + \left(1 - \rho_{\sigma_d}^2\right)^{\frac{1}{2}} \eta_d u_{dt}$$
where $u_{dt} \sim \mathcal{N}(0, 1)$

and

$$\log \sigma_{\varphi t} = \rho_{\sigma_\varphi} \log \sigma_{\varphi t-1} + \left(1 - \rho_{\sigma_\varphi}^2\right)^{\frac{1}{2}} \eta_\varphi u_{\varphi t}$$
where $u_{\varphi t} \sim \mathcal{N}(0, 1)$.

This parsimonious specification introduces only four new parameters, ρ_{σ_d}, ρ_{σ_φ}, η_d, and η_φ, while being surprisingly powerful in capturing important features of the data (Shephard, 2008). All the shocks and innovations throughout the model are observed by the agents when they are realized. Agents have, as well, rational expectations about how these shocks (and all the other shocks to the economy) evolve over time. We can interpret the shocks to preferences and to their volatility as reflecting the random evolution of more complicated phenomena, such as changing demographics (see Fernández-Villaverde and Rubio-Ramirez, 2008).

Although we assume complete financial markets, to ease notation we drop the Arrow securities implied by that assumption from the budget constraints (they are in net zero supply at the aggregate level). Households also hold b_{jt} government bonds that pay a nominal gross interest rate of R_{t-1}. Therefore, the jth household's budget constraint is given by:

$$c_{jt} + x_{jt} + \frac{m_{jt}}{p_t} + \frac{b_{jt+1}}{p_t} = w_{jt} l_{jt} + \left(r_t u_{jt} - \mu_t^{-1} \Phi\left[u_{jt}\right]\right) k_{jt-1} + \frac{m_{jt-1}}{p_t} + R_{t-1} \frac{b_{jt}}{p_t} + T_t + \digamma_t$$

where x_t is investment, w_{jt} is the real wage, r_t is the real rental price of capital, $u_{jt} > 0$ is the rate of use of capital, $\mu_t^{-1}\Phi\left[u_{jt}\right]$ is the cost of using capital at rate u_{jt} in terms of the final good, μ_t, is an investment-specific technological level, T_t is a lump-sum transfer, and $\digamma_t$ is firms' profits. We specify that $\Phi[u] = \Phi_1(u-1) + \frac{\Phi_2}{2}(u-1)^2$, a form that satisfies that $\Phi[1] = 0$, $\Phi'[\cdot] = 0$, and $\Phi''[\cdot] > 0$. This function carries the normalization that $u = 1$ in the balanced growth path of the economy. Using the relevant first-order conditions, we can find $\Phi_1 = \Phi'[1] = \widetilde{r}$ where $\widetilde{r}$ is the (rescaled) steady-state rental price of capital (determined by the other parameters in the model). This leaves us with only one free parameter, Φ_2.

Given a depreciation rate δ and an investment adjustment cost parameter κ, the capital accumulated by household j at the end of period t is given by:

$$k_{jt} = (1-\delta) k_{jt-1} + \mu_t \left(1 - \frac{\kappa}{2}\left(\frac{x_{jt}}{x_{jt-1}} - \Lambda_x\right)^2\right) x_{jt}.$$

This function is written in deviations with respect to the balanced growth rate of investment, Λ_x. The investment-specific technology level μ_t, follows a random walk in logs, $\log \mu_t = \Lambda_\mu + \log \mu_{t-1} + \sigma_\mu \sigma_{\mu t} \varepsilon_{\mu t}$ with $\varepsilon_{\mu t} \sim \mathcal{N}(0, 1)$ and where Λ_μ is the drift of the process and $\varepsilon_{\mu t}$ is the innovation to its growth rate. The standard deviation $\sigma_{\mu t}$ also evolves as:

$$\log \sigma_{\mu t} = \rho_{\sigma_\mu} \log \sigma_{\mu t-1} + \left(1 - \rho_{\sigma_\mu}^2\right)^{\frac{1}{2}} \eta_\mu u_{\mu t}$$
where $u_{\mu t} \sim \mathcal{N}(0, 1)$.

Again, we can interpret this stochastic volatility as a stand-in for a more detailed explanation of technological progress in capital production that we do not model explicitly.

Each household j supplies a different type of labor services l_{jt} that are aggregated by a labor packer into homogeneous labor l_t^d with the production function $l_t^d = \left(\int_0^1 l_{jt}^{\frac{\eta-1}{\eta}} dj\right)^{\frac{\eta}{\eta-1}}$ that is rented to intermediate good producers at wage w_t. The labor packer is perfectly competitive and it takes all wages as given. Households set their wages with a Calvo pricing mechanism. At the start of every period, a randomly selected fraction $1-\theta_w$ of households can reoptimize their wages (where, by a law of large numbers, individual probabilities and aggregate fractions are equal). All other households index their wages given past inflation with an indexation parameter $\chi_w \in [0, 1]$.

5.2. Firms

There is one final good producer that aggregates a continuum of intermediate goods according to:

$$y_t = \left(\int_0^1 y_{it}^{\frac{\varepsilon-1}{\varepsilon}} di\right)^{\frac{\varepsilon}{\varepsilon-1}} \tag{24}$$

where ε is the elasticity of substitution. The final good producer is perfectly competitive and minimizes its costs subject to the production function (24) and taking as given all intermediate goods prices p_{ti} and the final good price p_t.

Each intermediate good is produced by a monopolistic competitor with technology $y_{it} = A_t k_{it-1}^{\alpha}\left(l_{it}^d\right)^{1-\alpha}$, where k_{it-1} is the capital rented by the firm, l_{it}^d is the amount of the "packed" labor input rented by the firm, and A_t is neutral productivity. Productivity evolves as $\log A_t = \Lambda_A + \log A_{t-1} + \sigma_A \sigma_{At} \varepsilon_{At}$ where Λ_A is the drift of the process and $\varepsilon_{At} \sim \mathcal{N}(0, 1)$ is the innovation to its growth rate. The time-varying standard deviation of this innovation follows:

$$\log \sigma_{At} = \rho_{\sigma_A} \log \sigma_{At-1} + \left(1 - \rho_{\sigma_A}^2\right)^{\frac{1}{2}} \eta_A u_{At}$$
where $u_{At} \sim \mathcal{N}(0, 1)$.

Intermediate good producers meet the quantity demanded by the final good producer by renting l_{it}^d and k_{it-1} at prices w_t and r_t. Given their demand function, these producers set prices to maximize profits. However, when they do so, they follow a Calvo pricing scheme. In each period, a fraction $1-\theta_p$ of intermediate good producers reoptimize their prices. All other firms partially index their prices by past inflation with an indexation parameter $\chi \in [0, 1]$.

5.3. The monetary authority

A monetary authority sets the nominal interest rate R_t (as a deviation with respect to R, the balanced growth path nominal interest rate) by following a Taylor rule:

$$\frac{R_t}{R} = \left(\frac{R_{t-1}}{R}\right)^{\gamma_R} \left(\left(\frac{\Pi_t}{\Pi}\right)^{\gamma_\Pi \gamma_{\Pi,t}} \times \left(\frac{y_t^d}{y_{t-1}^d} / \exp\left(\Lambda_{y^d}\right)\right)^{\gamma_y \gamma_{y,t}} \right)^{1-\gamma_R} \xi_t. \tag{25}$$

The first term on the right-hand side represents a desire for interest rate smoothing. The second term responds to the deviation of inflation from its balanced growth path level Π. The third term is a "growth gap": the ratio between the growth rate of the economy and Λ_{y^d}, the balanced path gross growth rate of y_t^d, where y_t^d is aggregate demand (defined precisely in appendix 3). The last term is the monetary policy shock, where $\log \xi_t = \sigma_\xi \sigma_{\xi,t} \varepsilon_{\xi t}$ with an innovation $\varepsilon_{\xi t} \sim \mathcal{N}(0, 1)$ and a time-varying standard deviation, $\sigma_{\xi,t}$, that follows an autoregressive process

$$\log \sigma_{\xi t} = \rho_{\sigma_\xi} \log \sigma_{\xi t-1} + \left(1 - \rho_{\sigma_\xi}^2\right)^{\frac{1}{2}} \eta_\xi u_{\xi,t}$$

where $u_{\xi,t} \sim \mathcal{N}(0, 1)$.

In this policy rule, we have two drifting parameters: the responses of the monetary authority to the inflation gap and the growth gap. The parameters drift over time as:

$$\log \gamma_{\Pi t} = \rho_{\gamma_\Pi} \log \gamma_{\Pi t-1} + \sigma_\pi \varepsilon_{\pi t} \quad \text{and}$$
$$\log \gamma_{yt} = \rho_{\gamma_y} \log \gamma_{yt-1} + \sigma_y \varepsilon_{yt} \quad \text{where } \varepsilon_{\pi t}, \varepsilon_{yt} \sim \mathcal{N}(0, 1).$$

For simplicity, the volatility of the innovation to this processes is fixed over time. The agents perfectly observe these changes in monetary policy parameters.

5.4. Equilibrium and solution

We characterize the equilibrium of the model in appendix 3. The equilibrium conditions are non-stationary because we have two unit roots in the processes for technology. We circumvent this problem by rescaling the model using the variable $z_t = A_t^{\frac{1}{1-\alpha}} \mu_t^{\frac{\alpha}{1-\alpha}}$ in the form: $\widetilde{k}_t = \frac{k_t}{z_t \mu_t}$, $\widetilde{c}_t = \frac{c_t}{z_t}$, $\widetilde{x}_t = \frac{x_t}{z_t}$, $\widetilde{y}_t = \frac{y_t}{z_t}$, $\widetilde{w}_t = \frac{w_t}{z_t}$, $\widetilde{r}_t = \mu_t r_t$, $\widetilde{A}_t = \frac{A_t}{\exp(\Lambda_A) A_{t-1}}$, and $\widetilde{\mu}_t = \frac{\mu_t}{\exp(\Lambda_\mu)\mu_{t-1}}$. In the notation of Section 2.1 we have that:

1. The states of the (rescaled) economy are
$$\mathcal{S}_t = \left(\log \widetilde{k}_{t-1}, \log \widetilde{c}_{t-1}, \log \widetilde{x}_{t-1}, \log \widetilde{y}_{t-1}, \log \upsilon_{t-1}^p, \log \upsilon_{t-1}^w, \log \widetilde{w}_{t-1}, \log R_{t-1}, \log \Pi_{t-1}\right)'.$$
2. The structural shocks are $\mathcal{Z}_t = (\log d_t, \log \varphi_t, \log \widetilde{\mu}_t, \log \widetilde{A}_t, \log \xi_t, \log \gamma_{\Pi t}, \log \gamma_{yt})'$. The parameter drifts are handled as structural shocks in the state-space representation.
3. The volatility shocks are $\Sigma_t = (\log \sigma_{dt}, \log \sigma_{\varphi t}, \log \sigma_{\mu t}, \log \sigma_{At}, \log \sigma_{\xi t}, 0, 0)'$, where the last two zeros correspond to the processes for parameter drifting, which have constant volatilities (see also in vector $\mathcal{U}_t$ below). Here, we can see that we do not need as many volatility shocks (5 of them) as structural shocks (7 of them).
4. The innovations to the structural and the volatility shocks are $\mathcal{E}_t = (\varepsilon_{dt}, \varepsilon_{\varphi t}, \varepsilon_{\mu t}, \varepsilon_{At}, \varepsilon_{\xi t}, \varepsilon_{\pi t}, \varepsilon_{yt})'$ and $\mathcal{U}_t = (u_{dt}, u_{\varphi t}, u_{\mu t}, u_{At}, u_{\xi t}, 0, 0)'$, respectively.

We pick as observables the first difference of the log of the relative price of investment, the log federal funds rate, log inflation, the first difference of log output, and the first difference of log real wages, in our notation $\mathcal{Y}_t = \left(-\triangle \log \mu_t, \log R_t, \log \Pi_t, \triangle \log y_t, \triangle \log w_t\right)'$. We select these variables because they bring us information about aggregate behavior (output), the stance of monetary policy (the interest rate and inflation), and the different shocks (the relative price of investment about investment-specific technological change, the other four variables about technology and preference shocks) that we are concerned about. Note that we have the same number of observables as we do of volatility shocks, as required by Theorem 5.

Also, note that a second-order approximation is even more relevant in our application than in the standard case of dynamic equilibrium models with stochastic volatility because a linearization would also imply that the parameter drift in the Taylor rule would disappear as well from the equilibrium dynamics (see appendix 4 for more details).

5.5. A user's guide to computing the results

In this subsection, we outline a user's guide to the computation of our result. The main steps involved are:

1. We rescale the equilibrium conditions of the model to make them stationary and write them in `Mathematica`.
2. We ask `Mathematica` to take all the analytic derivatives required to solve for the second-order approximation of the model. We take advantage of the symbolic computation capabilities of `Mathematica` to express them as functions of parameters of the model. In that way, we do not need to recompute the derivatives, the most time-intensive step, for each set of parameter values in our estimation.
3. Once we have all the relevant derivatives, we export them into `Fortran` files. At this stage, we have a set of `Fortran` files that solves the second-order approximation of the dynamics of the model as a function of the parameters (steps 2 and 3 take about 3 h).
4. Then, we compile the resulting files with the `Intel Fortran Compiler` version 10.1.025 with IMSL. Compilation takes about 18 h. The project has 1798 files and occupies 2.33 Gbytes of memory.
5. Given some parameter values, we use the derivatives from step 3 to solve for the second-order approximation of the model. For this task, `Fortran` takes around 5 s (remember that we have 9 states, 7 structural shocks, and 5 volatility shocks).
6. We build, in `Fortran`, the state-space representation associated with the second-order approximation.
7. We approximate the likelihood with the particle filter using 10,000 particles. This number delivered a good compromise between accuracy and time to compute the likelihood. Hence, the solution of the model plus the evaluation of one likelihood requires 22 s on a Dell server with 8 processors.
8. We use steps 2–7 within a random-walk Metropolis–Hastings algorithm to draw from the posterior of the parameters. Drawing 5000 times from the posterior takes around 38 h. In order to initialize the chain, we extensively search on a grid parameter for high values of the likelihood.

The `Mathematica` and `Fortran` codes were highly optimized in order to (1) keep the size of the project within reasonable dimensions (otherwise, the compiler cannot parse the files) and (2) provide a rapid solution of the model and a fast computation of the likelihood. Perhaps the most important task in that optimization was the parallelization of the `Fortran` code using `OPENMP` as well as the compilation options: `OG` (global optimizations) and `Loop`

Table 5.1
Fixed parameters.

β	h	ψ	ϑ	δ	α	κ	ε	η	Φ_2	$\rho_{\gamma\Pi}$	$\rho_{\gamma y}$	σ_y
0.99	0.9	8	1.17	0.025	0.21	9.5	10	10	0.001	0.95	0	0

`Unroll`. Without the parallelization, the solution of the model and evaluation of its likelihood take about 70 s.

Also, note that implementing corollary 1 needed in step 7 requires the solution of a linear system of equations and the computation of a Jacobian. For our application, we found that the following sequence of LAPACK operations delivered the fastest solution:

1. DGESV (computes the solution to a real system of linear equations $A * X = B$).
2. DGETRI (computes the inverse of a matrix using the LU factorization from the previous line).
3. DGETRF (helps to compute the determinant of the inverse from the previous line).

With respect to the random-walk Metropolis–Hastings, we performed an intensive process of fine-tuning of the chain, both in terms of initial conditions as well as in terms of getting the right acceptance level. While the tuning was time-intensive, it did not involve any non-standard step. The only other important remark is to remember to keep the random numbers used for resampling in the particle filter constant across draws of the Markov chain. This is required to reduce the numerical variance of the procedure, which was a serious concern for us given the complexity of our problem.

5.6. Data and estimation

We estimate our model using the five time series for the US economy described above. Our sample covers 1959.Q1–2007.Q1, with 192 observations. We stop at 2007 to avoid having to deal with the financial crisis, which would make it difficult to appreciate the points we want to illustrate about how to econometrically deal with stochastic volatility. This could be fixed at the cost of a lengthier discussion. Appendix 5 explains how we construct the series.

Once we have evaluated the likelihood, we combine it with a prior. We pick flat priors on a bounded support for all the parameters. The bounds are either natural economic restrictions (for instance, the Calvo and indexation parameters lie between 0 and 1) or are so wide that the likelihood assigns (numerically) zero probability to values outside them. Bounded flat priors induce a proper posterior, a convenient feature for our exercises below. We resort to flat priors for two reasons. First, to reduce the impact of presample information and show that our results arise mainly from the shape of the likelihood and not from the prior (although, of course, flat priors are not invariant to reparameterization). Thus, we could interpret our posterior modes as maximum likelihood point estimates. Second, as we learned in Fernández-Villaverde et al. (2010b), eliciting priors for stochastic volatility is difficult, since we deal with unfamiliar units, such as the variance of volatility shocks, about which we do not have clear beliefs. Flat priors come, though, at a price: before proceeding to the estimation, we have to fix several parameters to reduce the dimensionality of the problem.

Table 5.1 lists the fixed parameters. Our guiding criterion in selecting them was to pick conventional values. The discount factor, $\beta = 0.99$, is a default choice, habit persistence, $h = 0.9$, matches the observed sluggish response of consumption to shocks, the parameter controlling the level of labor supply, $\psi = 8$, captures the average amount of hours in the data, and the depreciation rate, $\delta = 0.025$, induces the appropriate capital–output ratio. The elasticities of substitution, $\varepsilon = \eta = 10$, deliver average markups of around 10%, a common value in these models. We set the cost of capital utilization, Φ_2, to a small number to introduce some curvature in this decision. Three parameter values are borrowed from Fernández-Villaverde et al. (2009). The first is the inverse of the Frisch labor elasticity, $\vartheta = 1.17$. This aggregate elasticity is compatible with the micro data, once we allow for intensive and extensive margins on labor supply. The second is the coefficient of the intermediate goods production function, $\alpha = 0.21$. This value is lower than the common calibration in real business cycle models because, in our environment, we have positive profits that appear as capital income in the National Income and Product Accounts. Finally, the adjustment cost, $\kappa = 9.5$, is in line with other estimates from similar models (κ would be particularly hard to identify, since investment is not one of our observables).

Table 5.2
Posterior, parameters of nominal rigidities and structural shocks.

θ_p	χ	θ_w	χ_w	ρ_d	ρ_φ	Λ_μ	Λ_A
0.8139	0.6186	0.6869	0.6340	0.1182	0.9331	0.0034	0.0028
(0.0143)	(0.024)	(0.0432)	(0.0074)	(0.0049)	(0.0425)	(6.6e−5)	(4.1e−5)

The autoregressive parameter of the evolution of the response to inflation, $\rho_{\gamma\Pi}$, is set to 0.95. In preliminary estimations, we discovered that the likelihood pushed this parameter to 1. When this happened, the simulations became numerically unstable: after a series of positive innovations to $\log \gamma_{\Pi t}$, the reaction of nominal interest rates to inflation could be too tepid for too long. The 0.95 value seems to be the highest value of $\rho_{\gamma\Pi}$ such that the problem does not appear. The last two parameters, ρ_{γ_y} and σ_y, are equal to zero because, also in exploratory estimations, the likelihood favored values of $\sigma_y \approx 0$. Thus, we decided to forget about them and make $\gamma_{y,t} = 1$.

To find the posterior, we proceed as follows. First, we define a grid of parameter values and check for the regions of high posterior density by evaluating the likelihood function in each point of the grid. This is a time-consuming procedure, but it ensures that we are searching in the right zone of the parameter space. Once we have identified the global maximum in the grid, we initialize a random-walk Metropolis–Hastings algorithm from this point. After an extensive fine-tuning of the algorithm, we use 10,000 draws from the chain to compute posterior moments.

5.7. Result I: parameter estimates

Our first empirical result is the parameter estimates. To ease the discussion, we group them in different tables, one for each set of parameters dealing with related aspects of the model. In all cases, we report the mode of the posterior and the standard deviation in parenthesis below (in the interest of space, we do not include the whole histograms of the posterior).

Table 5.2 presents the results for the nominal rigidities and the stochastic processes for the structural shock parameters. Our estimates indicate an economy with substantial rigidities in prices, which are reoptimized roughly once every five quarters, and in wages, which are reoptimized approximately every three quarters. Moreover, since the standard deviations are small, there is enough information in the data about this result. At the same time, there is a fair amount of indexation, between 0.62 and 0.63, which brings a strong persistence of inflation. While it is tempting to compare

Table 5.3
Posterior, parameters of the stochastic processes for volatility shocks.

$\log\sigma_d$	$\log\sigma_\varphi$	$\log\sigma_\mu$	$\log\sigma_A$	$\log\sigma_\xi$
−1.9834 (0.0726)	−2.4983 (0.0917)	−6.0283 (0.1278)	−3.9013 (0.0745)	−6.000 (0.1471)
ρ_{σ_d}	ρ_{σ_φ}	ρ_{σ_μ}	ρ_{σ_a}	ρ_{σ_ξ}
0.9506 (0.0298)	0.1275 (0.0032)	0.7508 (0.035)	0.2411 (0.005)	0.8550 (0.0231)
η_d	η_φ	η_μ	η_a	η_ξ
0.1007 (0.0083)	2.8316 (0.0669)	0.3115 (0.006)	0.7720 (0.013)	0.5723 (0.0185)

Table 5.4
Posterior, policy parameters.

γ_R	$\log\gamma_y$	Π	$\log\gamma_\Pi$	σ_π
0.7855 (0.0162)	−1.4034 (0.0498)	1.0005 (0.0043)	0.0441 (0.0005)	0.145 (0.002)

our estimates with the micro evidence on the individual duration of prices, in our model all prices and wages change every quarter. That is why, to a naive observer, our economy would look like one displaying tremendous price flexibility.

We estimate a low persistence of the intertemporal preference shock and a high persistence of the intratemporal one. The low estimate of ρ_d produces the quick variations in marginal utilities of consumption that match output growth and inflation fluctuations. The intratemporal shock is persistent to account for long-lived movements in hours worked. We estimate mean growth rates of technology of 0.0034 (neutral) and 0.0028 (investment-specific). Those numbers give us an average growth of the economy of 0.44% per quarter (0.46 in the data). Technology shocks, in our model, are deviations with respect to these drifts. Thus, we estimate that A_t falls in only 8 of the 192 quarters in our sample (which roughly corresponds to the percentage of quarters where measured productivity falls in the data), even if we estimate negative innovations to neutral technology in 103 quarters.

The results for the parameters of the stochastic volatility processes appear in Table 5.3. In all cases, the ρ's and the η's are far away from zero: the likelihood strongly favors values where stochastic volatility plays an important role. The standard deviations of the innovations of the intertemporal preference shock and of the monetary policy shock are the most persistent, while the standard deviation of the innovation of the intratemporal preference shock is the least persistent. The standard deviation of the innovations of the volatility shock to the intratemporal preference shock, $\eta_\varphi = 2.8316$, is large: the model asks for fast changes in the size of movements in marginal utilities of leisure to reproduce the hours data.

In Table 5.4, we have the estimates of the policy parameters. The autoregressive component of the federal funds rate is high, 0.7855, although somewhat smaller than in estimations without parameter drift. The value of γ_y (0.24 in levels) is similar to other results in the literature and shows that the likelihood clearly likes parameter drifting, although with mild persistence. The estimated value of Π plus the correction on equilibrium inflation implied by second-order effects of the solution match the average inflation in the data. Also, these second-order effects complicate the introduction of time-variation in Π. The likelihood wants to match the moments of the ergodic distribution of inflation, not the level of Π, which is inflation along the balanced growth path. When we have non-linearities, the mean of that ergodic distribution may be far from Π. Thus, learning about Π is hard. Learning about a time-varying Π is even harder. Finally, the estimated value of γ_Π (1.045 in levels) guarantees local determinacy of the equilibrium even if $\gamma_{\Pi,t}$ is temporarily below 1 (see appendix 6.1 for details).

In appendix 6.2, we plot the impulse response functions of the model implied by our estimates. This exercise allows us to check that the estimates are sensible and that the behavior of the model is consistent with the behavior of related models in the literature. In appendix 6.3, we compare our model against an alternative version without parameter drifting but still with stochastic volatility. That is, we ask whether, once we have included stochastic volatility, it is still important to allow for changes in the monetary policy rule to account for the time-varying volatility of US aggregate data over the last several decades. The results show that, even after controlling for stochastic volatility, the data strongly prefer a specification where the monetary policy rule has changed over time. In appendix 6.4, we show that this finding does not imply that volatility shocks did not play an important role in the time-varying volatilities of US aggregate time series. In other words, both stochastic volatility and parameter drifting are key parts of a successful dynamic equilibrium model of the US economy.

5.8. Results II: smoothed shocks

Fig. 5.1 reports the log-deviations with respect to their means for the smoothed intertemporal, intratemporal, and monetary shocks and deviations of the growth rate of the investment and technological shocks with respect to their means (±2 standard deviations). We color different vertical bars to represent each of the periods at the Federal Reserve: the Martin years from the start of our sample in 1959 to the appointment of Burns in February 1970 (white), the Burns–Miller era (light blue), the Volcker interlude from August 1979 to August 1987 (gray), the Greenspan times (orange), and Bernanke's tenure from February 2006 (yellow).

We see in the top left panel of Fig. 5.1 that the intertemporal shock, $\log d_t$, is particularly high in the 1970s. This increases households' desire for current consumption (for instance, because of the entrance of baby boomers into adulthood). A higher aggregate demand triggers, in the model, the higher inflation observed in the data for those years. The shock has a dramatic drop in the second quarter of 1980. This is precisely the quarter in which the Carter administration invoked the Credit Control Act (March 14, 1980). Schreft (1990) documents that this measure caused turmoil in financial markets and, most likely, distorted intertemporal choices of households, which is reflected in the large negative innovation to $\log d_t$. The low values of $\log d_t$ in the 1990s with respect to the 1970s and 1980s eased the inflationary pressures in the economy.

The shock to the utility of leisure, $\log\varphi_t$, grows in the 1970s and falls in the 1980s to stabilize at a very low value in the 1990s. The likelihood wants to track, in this way, the path of average hours worked: low in the 1970s, increasing in the 1980s, and stabilizing in the 1990s. Higher hours also lower the marginal cost of firms (wages fall relative to the technology level). The reduction in marginal costs also helped to reduce inflation during Greenspan's tenure.

The evolution of the investment-specific technology, $\log\widetilde{\mu}_t$, shows a sharp drop after 1973 (when it is likely that energy-intensive capital goods suffered the consequences of the oil shocks in the form of economic obsolescence) and large positive realizations in the late 1990s (our model interprets the sustained boom of those years as the consequence of strong improvements in investment technology). These positive realizations were an additional help to contain inflation during the 1990s. In comparison, the neutral-technology shocks, $\log\widetilde{A}_t$, have been stable since 1959, with only a few big shocks at the end of the sample.

The evolution of the monetary policy shock, $\log\xi_t$, reveals large innovations in the early 1980s. This is due both to the fast change in policy brought about by Volcker and to the fact that a Taylor rule might not fully capture the dynamics of monetary policy during a period in which money growth targeting was attempted. Sims

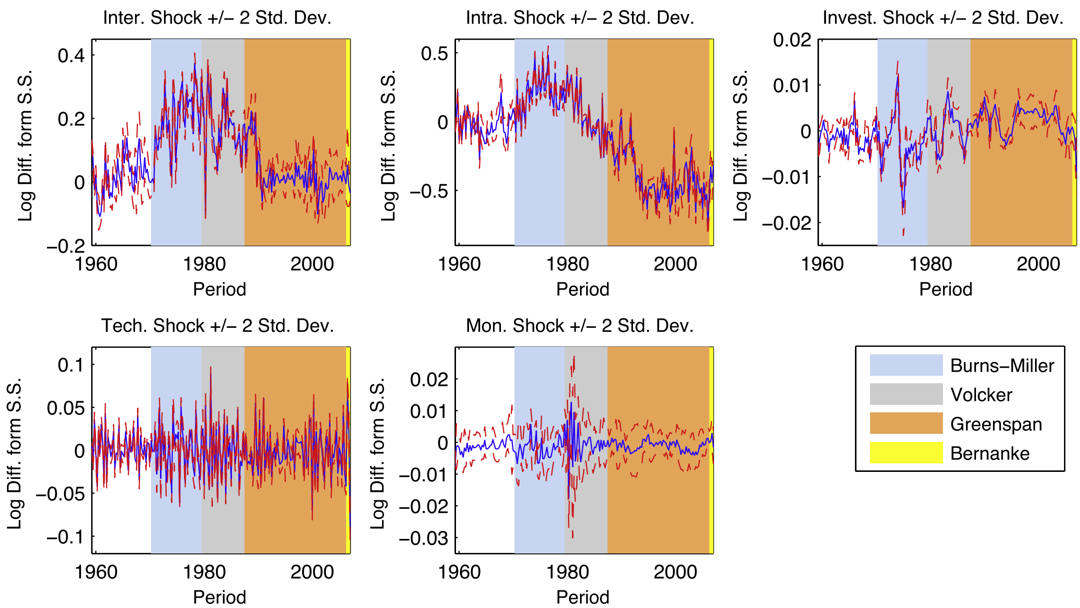

Fig. 5.1. Smoothed intertemporal ($\log d_t$) shock, intratemporal ($\log \varphi_t$) shock, investment-specific ($\log \widetilde{\mu}_t$) shock, technology ($\log \widetilde{A}_t$) shock, and monetary policy ($\log \xi_t$) shock +/−2 s.d. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

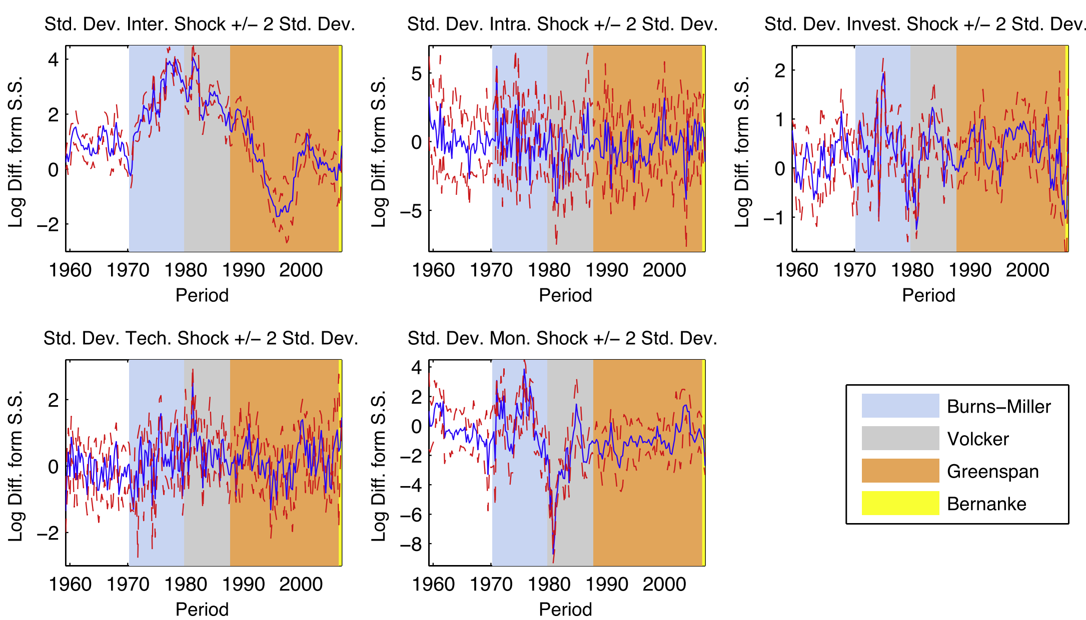

Fig. 5.2. Smoothed standard deviation shocks to the intertemporal ($\log \sigma_{dt}$) shock, the intratemporal ($\log \sigma_{\phi t}$) shock, the investment-specific ($\log \sigma_{\mu t}$) shock, the technology ($\log \sigma_{At}$) shock, and the monetary policy ($\log \sigma_{\xi t}$) shock +/−2 s.d.

and Zha (2006) also find that the Volcker period appears to be one with large disturbances to the policy rule and argue that the Taylor rule formalism can be a misleading perspective from which to view policy during that time. Our evidence from the estimated intertemporal, intratemporal, and investment shocks suggests that monetary authorities faced a more difficult environment in the 1970s and early 1980s than in the 1990s.

As a way to gauge the level of uncertainty of our smoothed estimates, we also plot in Fig. 5.1 the same shock (±2 standard deviations). In all cases, the data are informative about the history we just narrated.

We plot, in Fig. 5.2, the evolution of the volatility shocks, all of them in log-deviations with respect to their estimated means (plus/minus two standard deviations). We see in this figure that the standard deviation of the intertemporal shock was particularly high in the 1970s and only slowly went down during the 1980s and early 1990s. By the end of the sample, the standard deviation of the intertemporal shock was roughly at the level where it started. In comparison, the standard deviation of all the other shocks is relatively stable except, perhaps, for a large drop in the standard deviation of the monetary policy shock in the early 1980s as well as large changes in the standard deviation of the investment shock during the period of the oil price shocks. Hence, the 1970s and the 1980s were more volatile than the 1960s and the 1990s, creating a tougher environment for monetary policy. This result also confirms Blanchard and Simon's (2001) and Nason et al. (2008) observation that volatility had a downward trend in the 20th century with an abrupt and temporal increase in the 1970s. Also, from the

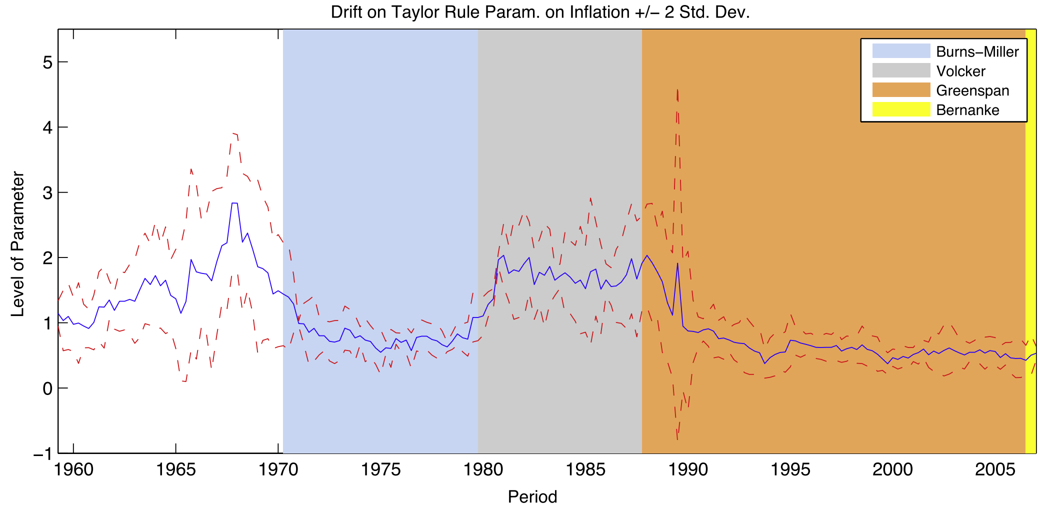

Fig. 5.3. Smoothed path for the Taylor rule parameter on inflation +/−2 standard deviations.

size of the plus/minus two standard deviations, we conclude that the big movements in the different series that we report can be ascertained with a reasonable degree of confidence.

Finally, in Fig. 5.3, we plot the evolution of the response of monetary policy to inflation plus/minus a two-standard-deviation interval. In particular, we graph $\gamma_{\Pi}\gamma_{\Pi t}$. This graph shows us an intriguing narrative. The parameter started the sample around its estimated mean, slightly over 1, and it grew more or less steadily during the 1960s until reaching a peak in early 1968. After that year, it suffered a fast collapse that took it below 1 in 1971. To put this evolution in perspective, it is useful to remember that Burns was appointed chairman in February 1970. The parameter stayed below 1 for all of the 1970s. The arrival of Volcker is quickly picked up by our smoothed estimates: it increases to over 2 after a few months and stays high during all the years of Volcker's tenure. Our estimate captures well the observation by Goodfriend and King (2007) that monetary policy tightened in the spring of 1980 as inflation and long-run inflation expectations continued to grow. Its level stayed roughly constant at this high during the remainder of Volcker's tenure. But as quickly as it rose when Volcker arrived, it went down again when he departed. Greenspan's tenure at the Fed meant that, by 1990, the response of the monetary authority to inflation was again below 1. Moreover, our estimates are relatively tight. Fernández-Villaverde et al. (2010a) discuss how the results of the estimation relate to historical evidence.

6. Conclusion

In this paper, we have shown how to estimate dynamic equilibrium models with stochastic volatility. The key to the procedure is to realize that a second-order perturbation to the solution of this class of models has a very particular structure that can be easily exploited to build an efficient particle filter. The recent boom in the literature on dynamic equilibrium models with stochastic volatility suggests that this procedure may have many uses. Our characterization of the solution might also be, on many occasions, of interest in itself to understand the dynamic properties of the equilibrium even if the researcher does not want to estimate the model.

As an application to illustrate how the procedure works we have estimated a business cycle model with both stochastic volatility in the structural shocks that drive the economy and parameter drifting in the monetary policy rule. Such a model is motivated by the need to have an empirical framework where we can account for the time-varying volatility of US aggregate time series. In particular, we have explained how you obtain point estimates in such a model and how to recover and analyze the smoothed structural and volatility shocks. Finally, through different comments – even if brief and not exhaustive – we have discussed the different empirical lessons that one can learn from all these steps.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at http://dx.doi.org/10.1016/j.jeconom.2014.08.010.

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